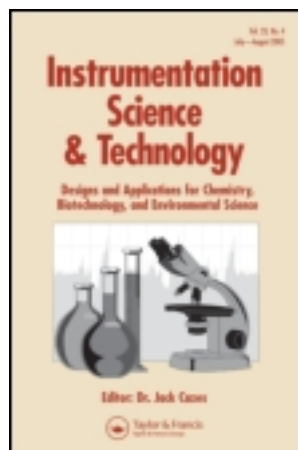


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CONSTRUCTION OF A PROTOTYPE LABORATORY-BUILT DESKTOP NMR SPECTROMETER

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□ A hardware system for a laboratory-built desktop nuclear magnetic resonance (NMR) spectrometer was constructed with a module design based on single-phase detection architecture. This spectrometer features an operating frequency of DC to 100 MHz, RF pulse power of 0.7244 mW to 1.2589 W, minimum detectable signal of 1.5136×10^{-10} mW, and receiver bandwidth of 20 kHz. Verification of the functional feasibility of NMR detection was implemented using a 30-mL custom probe in a 0.23-T magnetic field. A baseband free induction decay was also acquired through the application of an RF pulse at 1 W for 10 μ s. The observed initial amplitude and frequency of the MR signal are 1.47 V and 1.25 kHz, respectively. Meanwhile, the effective transverse relaxation time (T_2^*) for this configuration declines to no more than 15 ms because no shimming treatments were taken. The proposed desktop NMR spectrometer is suitable for building a hand-made NMR spectrometer in the laboratory, and enhances opportunities for novel and exciting NMR experiments.

Keywords FPGA programming, NMR spectrometer, RF signal process, single-phase detection

INTRODUCTION

Nuclear magnetic resonance (NMR) spectroscopy is a powerful analysis tool in the field of biochemistry. A unique feature of this technique is that it provides noninvasive detection in unknown samples. Ongoing advancements in communication technology and constant improvements in the performance/price ratio of radiofrequency (RF) devices continue to present numerous advantages in the field.^[1,2] However, most NMR spectrometers currently available are equipped with complex solid-state electronics, making for difficult comprehension of architecture and repair of problems occurring in the devices. In this study, a laboratory-built desktop

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NMR spectrometer was constructed, which is anticipated to present considerable advantages in education and research. The spectrometer can generate an RF signal with high accuracy and stability, as well as omitting the intermediate frequency circuit by basing an RF source on a direct digital synthesizer (DDS) chip. Building the spectrometer is easier because of single conversion, the carrier frequency of an RF signal directly translated to a baseband frequency (BF).^[3] A set of NMR experiments was implemented in a 0.23-T magnetic field of strength B_0 with a 30-mL sample of deionized water. The design verification was successful, also enabling the acquisition of the free induction decay (FID) of the sample tested.

In this work, the design for this spectrometer was first presented with several figures that were analyzed. The construction of the laboratory-built NMR spectrometer with respect to component selection is discussed in detail. The performance of the individual modules was systematically tested, followed by design verification. The feasibility of NMR detection as well as the shortcomings and future optimization required for the spectrometer are then discussed. Finally, we conclude that this prototype can be modified into a portable NMR spectroscopy after controlling and processing software is integrated with it.

DESIGN

A typical architecture for a pulse NMR spectrometer is represented in Figure 1. It can be divided into two chains, namely, the excitation and reception chains.

The broken lines indicate the programming and controlling signals for the RF generator and the TR switch, respectively. In this scheme, the single

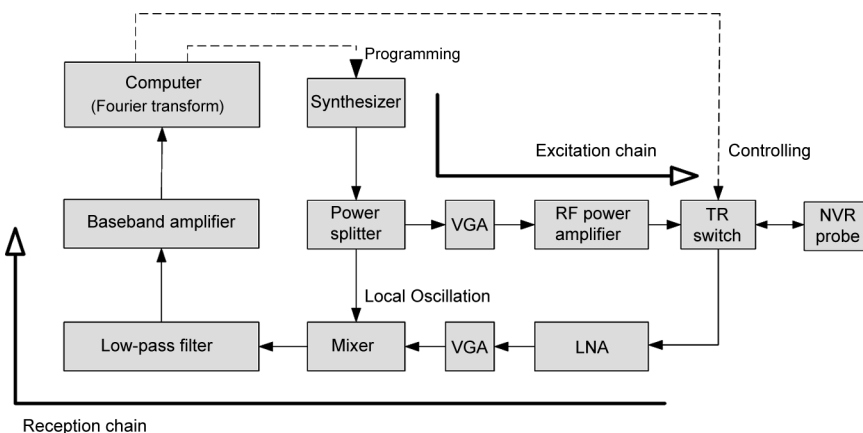


FIGURE 1 Pulsed NMR spectrometer system architecture.

coil in the probe is used for both spin excitation and detection relaxation. The spectrometer was designed according to several performance merits based on the principle of pulse NMR and RF transceiving,^[3] described in the succeeding subsections.

Operating Frequency Range

The Larmor angular frequency for certain types of nuclear signals, such as ^1H , can be expressed as follows:

$$\omega_0 = \frac{\gamma}{2\pi} \cdot B_0 \quad (1)$$

where γ is the gyromagnetic ratio, and $\gamma/2\pi = 42.577\text{MHz/T}$ for ^1H . The magnet currently available in our laboratory is a permanent 0.23-T magnet used for whole-body MR imaging; another 2-T magnet has a 5-cm gap. The operating frequency ranges from 0 to 100 MHz, which corresponds to 0–2.35 T.

RF Pulse Power

Upon interaction with the RF pulse, sample net magnetization $\vec{M}$ at equilibrium is tilted from its original position by an angle defined as follows:

$$\theta(r) = \gamma \cdot B_{1u}(r) \cdot I_p \cdot \tau \quad (2)$$

where r represents the position of a voxel in the sample, $B_{1u}(r)$ the alternating magnetic field generated by the pulse, I_p the peak value of the current $i(t) = I_p \sin(\omega t)$ following through the coil, of which ω is the Larmor angular frequency, and τ is the excitation pulse width. Assuming that all the active power (P) injected into the probe is dissipated in the coil series resistance (R_c), and the sample as well as $B_{1u}(r)$ are homogeneous, the appropriate power value to flip $\vec{M}$ by a given angle θ can be calculated using the equation^[4]

$$P = \frac{2R_c \cdot \theta^2}{\gamma^2 \cdot \tau^2 \cdot B_{1u}^2} = \frac{1}{2} R_c I_p^2 \quad (3)$$

RF pulse power has been implied as related to the geometry and electrical parameter of the coils used. A kind of solenoid probe (Vobs = 30 mL) was laboratory-built for the 0.23-T magnetic field, whose geometry and electrical parameters are summarized in Table 1.

TABLE 1 The Parameters of the Laboratory-Built Probe Coil

	Geometry				Electrical/9.835 MHz		
	Wire diameter	Coil length	Coil diameter	Turns	Resistance	Inductance	Q value
Values	2.68 mm	45 mm	34.32 mm	7	0.185 Ω	992.3 nH	320

For a custom transceiving coil, inserting a 30-mL serum bottle is a suitable approach. The electrical parameters are measured by an Agilent 4294A precision impedance analyzer. The magnetic field generated at the center of a solenoid can be calculated by^[5]

$$B = \frac{\mu_0}{4\pi} 2\pi nI (\cos \beta_1 - \cos \beta_2) \quad (4)$$

where μ_0 is the permeability in vacuum, n the turns per unit length, I the current feed, and β_1 and β_2 the characteristic angles of the coil. The power required is simulated using the aforementioned conditions as bases (Table 2).

A 50-ohm impedance is used in the calculations. Considering the available performance/price ratio of the RF power amplifier, the spectrometer was capable of outputting up to 1.5 W.

Noise Figure

The noise figure (F) of a receiver is a measure of by how much the signal-to-noise ratio (SNR) of the input signal is degraded. For a receiver comprising multiple two-port components in a cascade arrangement, the overall F can be computed as follows^[6]:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 \cdot G_2} + \cdots + \frac{F_n - 1}{G_1 \cdot G_2 \cdots G_{n-1}} \quad (5)$$

where $F_1, \dots, F_n$ are the noise figures for each of the components, and $G_1, \dots, G_n$ are the gains. The noise figure of the foremost stage in receiver

TABLE 2 The Required Power for the Laboratory-Built Probes

	Duration	
	10 μ s	100 μ s
Power	1 320 mW/31.2 dBm	13.2 mW/11.2 dBm

F_1 is dominant in F . The commercial spectrometers often have a noise figure below 1 dB. The main requirement of the laboratory-built spectrometer is to acquire abrupt FID signals. This specification is made less stringent (e.g., 5 dB) considering the cost of high-quality low-noise amplifiers.^[3]

Detection Bandwidth

The detection bandwidth for the spectrometer is determined by the FID signal being tested. Furthermore, the bandwidth of the detected signal is determined by the pulse sequence used. For ^1H -NMR spectroscopy, the typical bandwidth is 5 kHz.^[4] Thus, the detection bandwidth of the proposed spectrometer is designed to be 5 kHz. Because exciting the sample at a frequency that precisely matches the resonance frequency results in a DC signal at the output of the mixer, the excitation frequency is slightly offset from the range of precession frequencies to be examined by a value known as the spectral width (SW). Consequently, this imposes a constraint on pulse width τ that should satisfy the relationship $\tau \leq 1/(SW)$.

CONSTRUCTION AND TESTING

The spectrometer is based on the receiving theory of single conversion, which involves only one mixer. For simplification, the MR signal in the prototype is detected by the single-phase sensitive method, that is, single-phase sensitive detection.

Construction

The scheme design is described in detail as follows. An RF source based on a DDS chip is employed to generate continuous-wave (CW) signals for spin excitation. To obtain a more intensive local oscillator (LO) signal, as well as lessen the burden of the power amplifier (PA), a source amplifier is inserted before power splitting. Considering isolation with the rear power leveling circuits in idle condition, two RF switches that are controlled by a field-programmable gate array (FPGA) chip clocked at 40 MHz are used. An attenuator up to 31.5 dB with a minimum step of 0.5 dB is used to protect the PA effectively according to a 1-dB compression point. To connect the RF coil as a transceiver, a passive transmit/receive (TR) switch based on a diode pair interfaces with the PA, the probe, and the receiving circuits. Two low-noise amplifiers (LNA) are employed for the application of weak NMR signals involved, such as mass-limited samples or low fields. After

TABLE 3 The Primary Components for the Hardware of the Laboratory-Built NMR Spectrometer

Module	Device	Description
RF source		
Synthesizer	AD9852	Current output, DC–120 MHz
Amplifier	ZFL-500LN	50 Ω , gain: 24 dB, output 1 dB compr.: +5 dBm, 0.1–500 MHz
Single-chip proc.	Atmega16-8 L	8-bit microcontroller, 32 $\times$ 8 work registers, on-chip 2-cycle mult.
Power splitting		
RF splitter	ZMSC-2-1	50 Ω , 2 way out with 0° difference, 0.1 to 400 MHz
Pulse modulation		
RF switches	ZYSW-2-50DR	50 Ω , TTL driver, fast rise/fall time of 6 ns, DC to 5 GHz
TTL control	XC3S400	Equivalent logic cells: 8064, system clock: 40 MHz
Power adjust		
Attenuator	GKTS2-6-31.5-1	2 W, step length: 0.5, 1, 2, 4, 8 and 16, DC–1 GHz
Power amp.	ZHL-3A	50 Ω , gain: 25 dB, output 1 dB compr.: +30 dBm, 0.4–150 MHz
TR switch		
Diode Pair	BAV99	V _F : 855 mV with 10 mA, T _{RR} : 6 ns
Preamplification		
Low Noise Amps.	ZFL-500LN	50 Ω , gain: 24 dB, output 1 dB compr.: +5 dBm, 0.1–500 MHz
Downconversion		
Mixer	ZAD-1+	50 Ω , LO Power: +7 dBm, 0.5 to 500 MHz
Baseband adjust		
Amplifier	OPA228	Op-amp, low noise: 3 nV/, bandwidth: 8 MHz, 2.3 V/ μ s
Low-pass Filter	OPA2228	CMRR: 138 dB, Open-loop gain: 160 dB
Human–machine interface		
Personal Comp.	Satellite 4600	Pentium III 850 MHz, 128 M SDRAM

channel open. The 8648B can output sine waves ranging from 9 kHz to 2000 MHz with powers up to 10 mW. The tests for each module in the spectrometer are discussed in the following subsections.

TABLE 4 The Requirements of the Supplies in the Laboratory-Built Spectrometer

Voltage Level	Statistics	Sum
+24 V	ZHL-3A: max 600 mA	600 mA
$\pm$ 5 V	ZYSW-2-50DR: 40 mA, baseband circuit: 75 mA	2 $\times$ 40 + 75 = 155 mA
+15 V	ZFL-500LN: 60 mA	3 $\times$ 60 = 180 mA
+5 V	AD9852: max 922 mA, Xilinx 3S400: max 800 mA	1 722 mA



FIGURE 3 The inside and outside of the laboratory-built spectrometer.

RF Excitation Chain

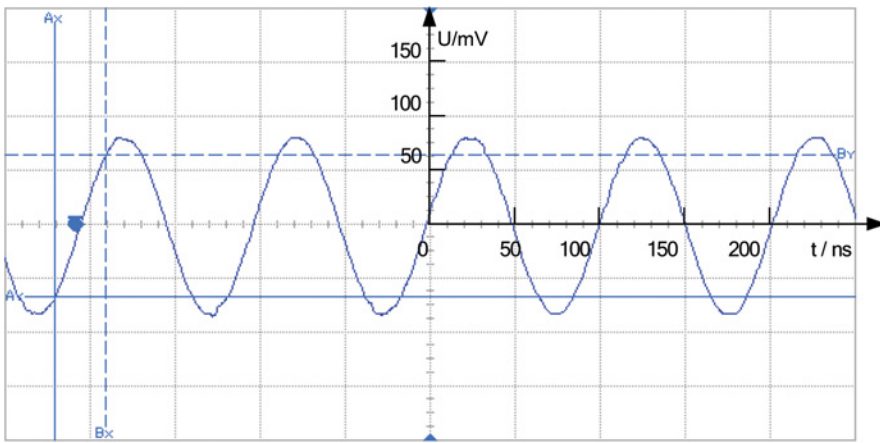
- RF source: The AD9852 chip is programmed under a single-chip processor that is ATmega16-8LE compliant in 3.3V standard. For the stability of the output frequency, a surface-mounted temperature-compensated crystal oscillator of 40 MHz with a temperature coefficient of 1.5 ppm was selected as the reference clock supplier (SMD0532-40S, ZGC. Electronic Co., Ltd). According to the data sheet, the output frequency can be determined as follows^[7]:

$$FTW = \frac{\text{Desired Output frequency} \times 2^N}{SYSCLK} \quad (6)$$

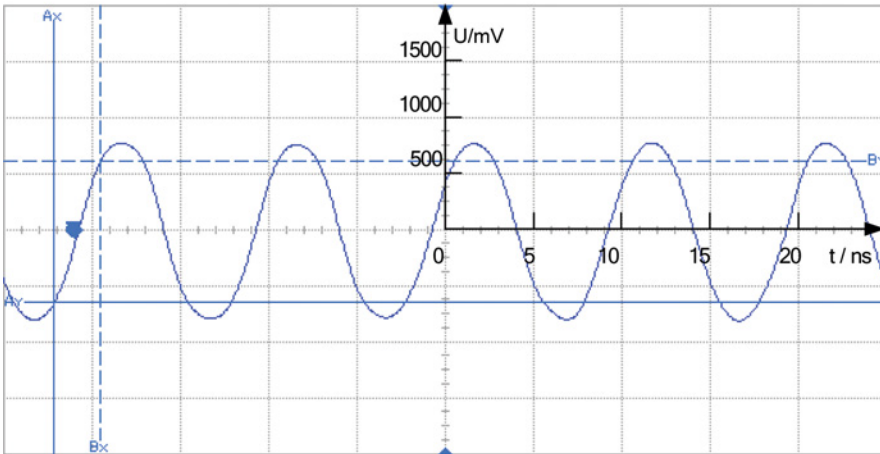
where FTW is the frequency tuning word interpreted by the synthesizer, and N and $SYSCLK$ are the phase accumulator resolution (48 bits in this study) and the working clock used, respectively. Two examples of continuous wave output are shown in Figure 4.

Note that the signals demonstrated in Figure 4 are the amplified output of the AD9852 chip after power promotion. Thus, the converted power of these signals is up to 5.4576 mW with the inverse sinc filter turned on. Under the same condition and having the output pass through a power splitter, the tested power for one output port is 2.75 mW. Meanwhile, the maximum nominal input for ZMSC-2-1 is 1 W.

- Pulse modulation and power amplifier: A proper connection scheme was determined based on the electrical schematic of high-speed switches used in the pulse modulation (PM). The FPGA chip, which was set to master serial mode, was programmed with the files preloaded in the on-board PROM as soon as the power is turned on. Moreover, the effect of the initial CW signal on the sample under detection can be avoided by



(a)



(b)

FIGURE 4 The sinewave outputs of RF source: (a) 9.835 MHz and (b) 100 MHz.

using two switches in tandem. One cycle time of the XC3S400 is $0.025 \mu\text{s}$, so that a $10\text{-}\mu\text{s}$ duration can be accomplished by setting the counter to count from 1 to 400. Figure 5 shows the captured single pulse after PM.

The oscilloscope was set in single trigger mode. After PM, the power of the signal decreases to about 1.9953 mW .

To substantially excite the sample under detection, the weak pulse signal of 2 mW needs further amplification. The corresponding input of the

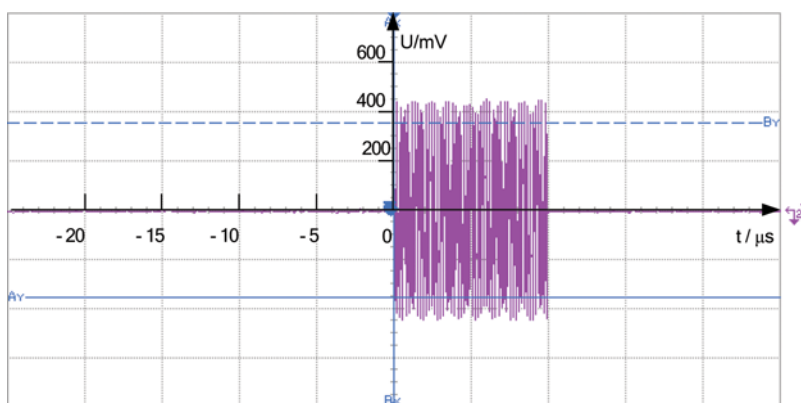


FIGURE 5 The square-shaped RF waves used for sample excitation.

1-dB compression point for the ZHL-3A is 3.1623 mW, which is obviously larger than 1.9953 mW. Despite increasing the dynamic range of the excitation, a stepping attenuator was incorporated into the samples with small quantities. Figure 6 shows the output of the PA with zero and complete attenuation, respectively.

From this measurement, the maximum output of the power adjustment module is 1 W, and the actual dynamic range is 31.4 dB.

RF Reception Chain

The reception chain for the laboratory-built spectrometer was also been tested. A setup was used to simulate single-phase detection in CW. The Agilent 8648B generator supplied the RF port signal, while the AD9852 outputted a signal to the LO port with a frequency difference of 1 kHz. Figure 7 shows the block diagram for this apparatus.

The diagram shows that the benefit for this scheme is that the entire reception chain, as well as parts of the excitation chain, can be tested in a single period. The ZAD-1 + RF mixer features 0.5–500 MHz with a normal local oscillation of 5.0119 mW. Therefore, the output of the 8648B generator was precisely adjusted according to the gain for ZFL-500LN to provide a modest signal at the LO port. The VGA OPA228 with a trimmer resistor was adjusted to provide a unit gain to the input signal. Shown in Figures 8a and 8b are the detected audiofrequency signals with and without low filtering to illustrate the function of demodulation and filtering, respectively.

The expected 1-kHz signal in Figure 8b is well detected after removing the unwanted harmonics and distortion. Thus, the reception function for the laboratory-built NMR spectrometer is feasible.

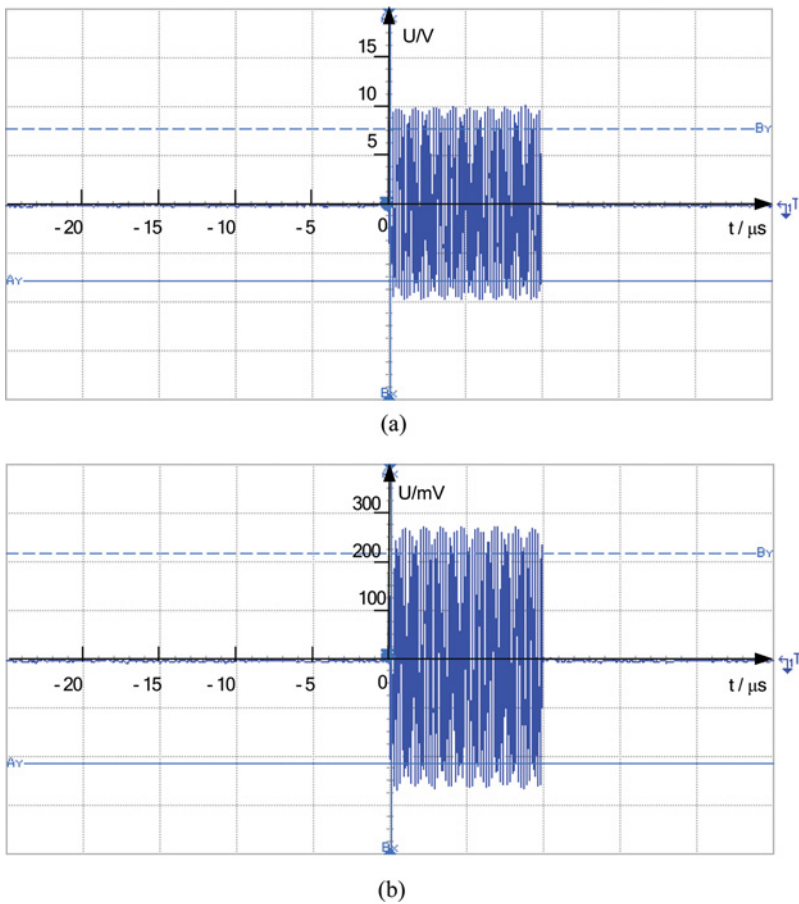


FIGURE 6. The tested outputs of the power amplifier following the attenuator (a) with zero attenuation and (b) with complete attenuation.

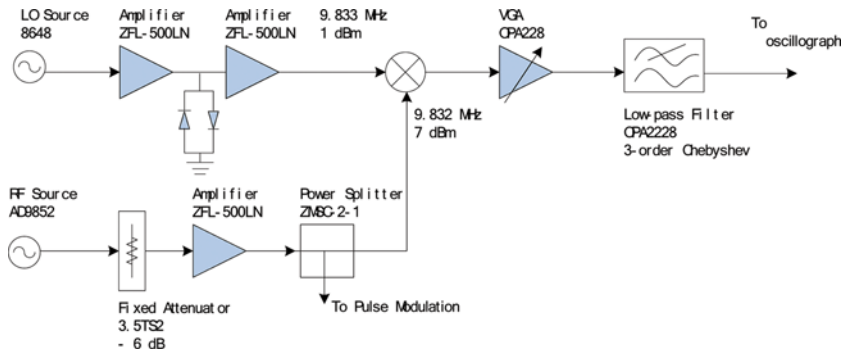


FIGURE 7 The diagram for the test of the receiver part of the laboratory-built spectrometer.

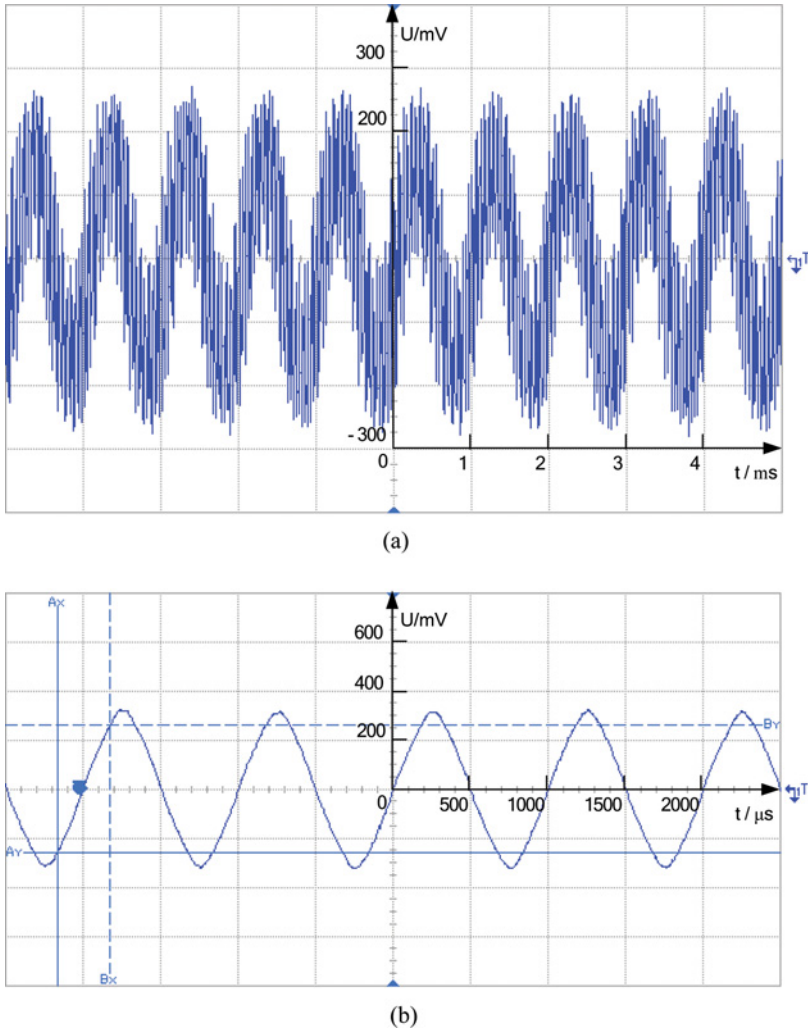


FIGURE 8 The outputs for the transceiver part of the spectrometer for basic function test: (a) without low filtering and (b) with low filtering.

Design Verification

To investigate the degree of practical performance in accordance with the individual design merits discussed earlier, we performed the verification of the prototype in terms of the aspects described as follows.

Operating Frequency Range

The present programming configuration for the AD9852 is at an internal working clock of 240 MHz. According to the Nyquist criterion,^[11]

therefore, the maximum output frequency of the DDS is 120 MHz in theory. The allowable output frequency is limited to 100 MHz considering certain practical matters, such as waveform smoothness. A signal of 99.71 MHz is well achieved. For excitation, the FTW can be calculated by employing all of the 48 bits of the output frequency of AD9852.

RF Excitation Power

The practical RF excitation power is controlled by the step attenuator that precedes the ZHL-3A PA. The test dynamic range for the RF pulse is 31.4 dBm, i.e., the range of the step attenuator. The exact output from the RF excitation chain can reach up to 1.4454 W when zero attenuation occurs. Also, the normal $P_{1\text{dB}}$ of the ZHL-3A is 1.5849 W under a 24-V supply. We conclude that the design merits for power excitation are well matched.

Noise Figure

Initially, the practical whole gain for the serial LNAs with the bandpass filters at the input of ZFL-500LN was tested with a power of 1×10^{-8} mW to investigate the noise performance of the receiver along the front part of the spectrometer. The calculated power for this signal is about 0.0158 mW under the 50-ohm impedance; thus, the actual gain for the LNAs in tandem is 62 dB. Meanwhile, using the same oscillator, the output noise of the second ZFL-500LN with no input at the first ZFL-500LN was also investigated. After converting the V_{pp} value into the rms power of the noise tested, 2.3988×10^{-4} mW was achieved. On the basis of 62 dB tested gain of the LNAs, we conclude that the minimum detectable signal in this system should be 1.5136×10^{-10} mW (e.g., the desired FID should be larger than this value).

Detection Bandwidth

As stated earlier, the detection bandwidth for the spectrometer is determined by the filter used for the baseband MR signal. In this hardware, the audiofrequency amplifier and filter were respectively constructed with a V/V magnification of 11 and a low-pass cutoff frequency of 20 kHz using the OPA2228 from Texas Instruments. The filter surpasses the single-tone test signal of 3.35 Vpp output for about 2.39 Vpp at the cutoff frequency of 20 kHz, whereas the output level significantly decreases to 154.8 mVpp when the input frequency increases to 30 kHz. Thus, the practical -3 dB bandwidth for the low-pass filter used is 20 kHz, which is in good agreement with the design merit.

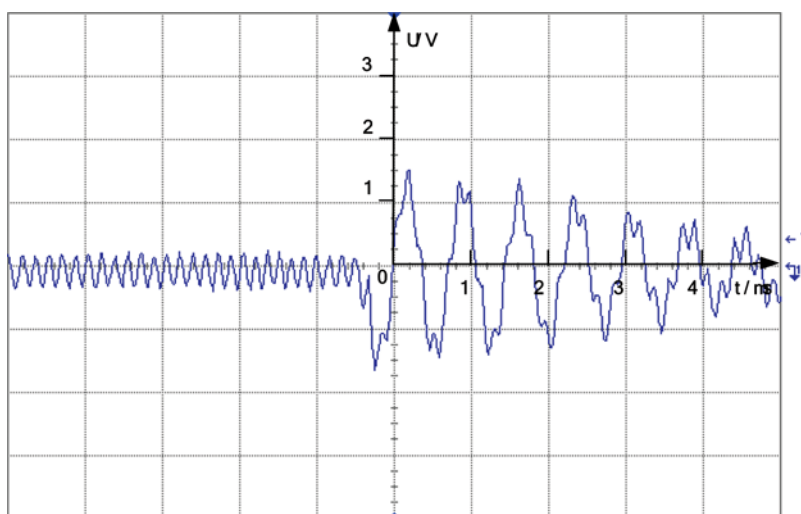


FIGURE 9 The demodulated MR signal appearing with fine oscillation decay.

NMR EXPERIMENTS AND DISCUSSION

An NMR experiment was implemented in a permanent 0.23-T magnet within the laboratory-built spectrometer. Several time-domain MR FID signals were well received at the output of the baseband adjustment circuit via the solenoid probe of the transceiver. To tune and match the solenoid with the sample loaded to 9.835 MHz under 50 ohms, the capacitances of C_t and C_m were respectively chosen to be 247.85 pF and 16 pF. The aqueous sample employed was deionized water of about 30 mL aligned vertically to B_0 in the magnet. Figure 9 shows an FID signal acquired using an RF pulse of 1 W within the duration of 10 μ s.

The resolution of the time axis was set to 1 ms/div with the longitudinal axis set to 1 V/div to conveniently observe the period and amplitude of the signal. The initial amplitude of the FID was determined to be robustly amplified to 1.48 V. Additionally, the background noise floor is so strong that it should not be neglected. The effective transverse relaxation time (T_2^*) will be no longer than 15 ms without the proposed shimming treatments.

DISCUSSION

Although the ability of MR signal detection was well illustrated by the laboratory-built NMR spectrometer, the time-domain signal should be further converted into a frequency domain via Fourier transform to obtain

the chemical information of the sample under analysis.^[12] One of the most commonly used methods is employing a digital card with enough sampling rate and resolution. In particular, a digitizer that can properly digitize the full bandwidth of the MR signal must be selected. For the general bandwidth of the ^1H spectrum and the reception bandwidth of the laboratory-built spectrometer, the requirement for the analog-to-digital rate should be above 50 ksp/s. Furthermore, the external trigger feature is an advantage if a precise control at the beginning of FID sampling after the end of the RF excitation pulse is considered.

Using the programming feature of the AD9852, the RF source in this spectrometer could output amplitude-modulated pulses, apart from hard pulses that have arbitrary shapes, by replacing the single-chip processor with a high-speed FPGA, such as the single sinc soft pulse.^[13] The FPGA would expand the application significantly for a selective excitation for a particular nuclear signal with more complex pulse sequence.

Further improvements should focus on reducing the amplitude of coupled pulses at the receiving port of the interface circuit when pulse excitation occurs. For the experimental condition of B_0 at a 0.23-T magnetic field and the particular coaxial cable used, the more practical $\lambda/4$ is 5.112 m, compared with that of 7.62 m in vacuum.^[14] Deviating from theory, the actual quantity of the signal monitored at the reception port is still sufficient under testing. To better understand the decoupling effect of the $\lambda/4$ line and the diode pair, a test pulse of 1 W was injected into the transmitting port of the interface circuit. In addition, powers of 549.54 mW and 0.7244 mW/539 mV_{pp} were received at the coil and reception ports, respectively. To further reduce the coupled voltage at the reception port, adding the numbers of cross-diode pairs in a shunt connection can effectively reduce impedance in the “diode-on” state.

CONCLUSION

A hardware system for a laboratory-built NMR spectrometer was designed and constructed according to several primary merits. This spectrometer aims to enhance the minimum spectrometer system in chemical analysis. Moreover, the design is anticipated to present portable applications because of its use of a low-field permanent magnet. In this paper, the implementation of a custom NMR spectrometer was described in detail from system requirements to system design and verification. With a DDS chip AD9852 as the RF source, the operating frequency measured ranges from DC to 100 MHz in accordance with the magnetic induction that ranges from 0 to 2.35 T. The nominal dynamic range for the RF excitation pulse is 31.4 dB as realized by a step attenuator that is in series with a

ZHL-3A PA, which features a maximum output of 1.5849 W. The receiver chain in the spectrometer was characterized with a minimum detectable signal of 1.5136×10^{-10} mW and the -3 dB bandwidth of 20 kHz. Functional feasibility for detecting the NMR FID was demonstrated successfully by imposing an RF pulse of 1 W at 10 μ s to a 30-mL deionized water sample. The preliminary result for the duration of the acquired MR signal is about 14 ms with a displayed dead time of 24 μ s without shimming.

Desktop NMR systems, such as the one presented in this paper, have the potential to substantially extend user-based MR spectroscopy. The inexpensive RF and digital components of the PC-based instrumentation have enabled the development of extremely low-cost and compact NMR systems that are of high quality. In parallel to the development of this laboratory-built spectrometer hardware, further software design for control and data processing, as well as its integration with the magnet and probe, is ongoing.

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REFERENCES

1. Giannini, V.; Craninckx, J.; Baschiroto, A. *Baseband Analog Circuits for Software Defined Radio*; Springer Netherlands: Dordrecht, The Netherlands, 2008.
2. Liu, C. J.; Huang, K. M.; Yan, L. P. *Design of Radio Frequency Circuits for Communication [B]*; Beijing Science Press: Beijing, China, 2005.
3. Brown, D. G. *Instrumentation for Parallel Magnetic Resonance Imaging [D]*; Ph.D. dissertation, Texas A&M University, 2005.
4. Massin, C. *Microfabricated Planar Coils in Nuclear Magnetic Resonance*; Hartung-Gorre Verlag: Konstanz, Germany, 2004; pp. 30–31.
5. Zhao, K. H.; Chen, X. M. *Electromagnetism* (2nd ed.); Hi Education Press: Beijing, China, 1985; pp. 361–367 (in Chinese).
6. Chang, K. *Microwave Solid-State Circuits and Applications*; Wiley: New York, 1994.
7. Analog Website. Available from: <http://www.analog.com>.
8. Minicircuits Website. Available from: <http://www.minicircuits.com>.
9. Texas Instruments Website. Available from: <http://www.ti.com>.
10. Toshiba Website. Available from: <http://www.toshiba.com/tai>.
11. Nyquist, H. Certain Topics in Telegraph Transmission Theory. *Trans. Am. Inst. Electrical Eng.* **1928**, *47*, 617–644.
12. Derome, A. E. *Modern NMR Techniques for Chemistry Research*; Pergamon Press: New York, 1987; p. 129.
13. Liu, Y.; Li, J. Q.; Li, G. Y. A Novel Selective-Excitation RF Pulse for Magnetic Resonance Imaging. *Magn. Reson. Mater. Phys.* **2008**, *21*(3), 199–205.
14. Bart, J.; Janssen, J. W. G.; van Bentum, P. J. M.; Kentgens, A. P. M.; Gardeniers, J. G. E. Optimization of Stripline-Based Microfluidic Chips for High-Resolution NMR. *J. Magnetic Resonance* **2009**, *201*, 175–185.